

Elementary Problems

Polynomial

Version 1.0

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Basic Work

1. Solve the equation $2x^2 + 7x + 3 = 0$ in real numbers.
2. Solve the equation $3x^2 - 2x + 6 = 0$ in real numbers.
3. Find all real numbers a such that the equation $ax^2 - 4x + a = 0$ has two distinct real solutions.
4. Solve the equation $x^3 + 5x^2 + 8x + 4 = 0$ in real numbers.
5. Solve the equation $2x^3 - \sqrt{3}x^2 - x - 2\sqrt{3} = 0$ in real numbers.
6. Let a be a real number. Find the range of a for which the equation

$$x^3 + (a+1)x^2 + (2a-4)x + (2-3a) = 0$$

has 3 distinct real roots.

7. Solve the equation $x^4 + x^3 - 10x^2 - 4x + 24 = 0$ in real numbers.
8. Solve the equation $x^4 + x^3 + x^2 + 2 = 0$ in real numbers.
9. Solve the equation $x^4 - 2x^3 - 22x^2 - 2x + 1 = 0$ in real numbers.
10. Solve the equation $x^4 - x^3 - 4x^2 + x + 1 = 0$ in real numbers.
11. Prove that a non-constant polynomial is unbounded.
12. Find the quadratic polynomial $P(x)$ such that $P(1) = 2$, $P(2) = 7$ and $P(3) = -1$.
13. Let $x_0, x_1, \dots, x_n$ be distinct rational numbers and let $y_0, y_1, \dots, y_n$ be rational numbers. If $P(x)$ is a degree n polynomial such that $P(x_j) = y_j$ for $j = 0, 1, \dots, n$, prove that the coefficients of $P(x)$ are rational numbers.
14. Let $P(x)$ be a polynomial with real coefficients. For any irrational number α , the number $P(\alpha)$ is irrational. Prove that the coefficients of $P(x)$ are rational numbers.
15. (USAMO 1975 Problem 3) If $P(x)$ denotes a polynomial of degree n such that

$$P(k) = \frac{k}{k+1}$$

for $k = 0, 1, 2, \dots, n$, determine $P(n+1)$.

16. Find the polynomial $P(x)$ of degree n such that $P(k) = 2^k$ for $k = 0, 1, \dots, n$.

17. (IMO shortlist 1981) Let P be a polynomial of degree n satisfying

$$P(k) = \binom{n+1}{k}^{-1} \quad \text{for } k = 0, 1, \dots, n.$$

Determine $P(n+1)$.

18. Let $P(x)$ be a monic polynomial of degree n . Prove that at least one of the numbers $|P(0)|, |P(1)|, \dots, |P(n)|$ is not less than $\frac{n!}{2^n}$.
19. (IMO shortlist 1968) A polynomial $p(x) = a_0x^k + a_1x^{k-1} + \dots + a_k$ with integer coefficients is said to be divisible by an integer m if $p(x)$ is divisible by m for all integers x . Prove that if $p(x)$ is divisible by m , then $k!a_0$ is also divisible by m . Also prove that if a_0, k, m are nonnegative integers for which $k!a_0$ is divisible by m , there exists a polynomial $p(x) = a_0x^k + \dots + a_k$ divisible by m .
20. Find all polynomials $P(x)$ such that $(x+2)P(x) = xP(x+2)$.
21. Find all polynomials $P(x)$ such that $P(x-1) + P(x+1) = 2P(x) + 2$.
22. Find all polynomials $P(x)$ such that $P(P(x)) = P(x^2)$.
23. Find all polynomials $P(x)$ and $Q(x)$ such that $P(x) = Q(x)(P(x) + Q(x))$.
24. (IMO shortlist 1967) Find all real solutions of the system of equations:

$$\sum_{k=1}^n x_k^i = a^i$$

for $i = 1, 2, \dots, n$.

25. Prove or disprove the following.

- (a) For any nonzero polynomials $P(x)$ and $Q(x)$, we have

$$\deg(P+Q) = \deg P + \deg Q.$$

- (b) For any nonzero polynomials $P(x)$ and $Q(x)$, we have

$$\deg(P+Q) = \max\{\deg P, \deg Q\}.$$

- (c) For any nonzero polynomials $P(x)$ and $Q(x)$, we have

$$\deg(PQ) = \deg P + \deg Q.$$

- (d) For any nonzero polynomial $P(x)$, we have $\deg P(x^2) = (\deg P)^2$.

- (e) The function $f(x) = |x|$ is not a polynomial.
- (f) Let $n \geq 1$ be an integer. Given any distinct real numbers $x_1, \dots, x_n$ and any real numbers $y_1, \dots, y_n$, there exists a unique polynomial $P(x)$ of degree n such that $P(x_j) = y_j$ for each $j = 1, 2, \dots, n$.

Identity

26. Factorize the following into polynomials with integer coefficients.

- (a) $a^2 - b^2$;
- (b) $a^3 \pm b^3$;
- (c) $a^n - b^n$ for positive integer n ;
- (d) $a^n + b^n$ for odd positive integer n ;
- (e) $a^3 + b^3 + c^3 - 3abc$;
- (f) $a^4 + 4b^4$;
- (g) $(ab + 1)^2 + (a - b)^2$;
- (h) $a^3b + b^3c + c^3a - ab^3 - bc^3 - ca^3$;
- (i) $(a + 1)(a + 2)(a + 3)(a + 4) + 1$.

27. Express the following in terms of the symmetric polynomials

$$S_1 = a + b + c, \quad S_2 = ab + bc + ca, \quad S_3 = abc.$$

- (a) $a^2 + b^2 + c^2$;
- (b) $a^3 + b^3 + c^3$;
- (c) $a^2b + a^2c + b^2c + b^2a + c^2a + c^2b$;
- (d) $a^2bc + b^2ca + c^2ab$;
- (e) $a^2b^2 + b^2c^2 + c^2a^2$;
- (f) $a^3b + a^3c + b^3c + b^3a + c^3a + c^3b$;
- (g) $a^4 + b^4 + c^4$.

28. Find all real numbers a, b, c such that $a^2 + b^2 + c^2 = ab + bc + ca$.

29. Let a, b, c be real numbers such that $a + b + c = 0$ and $a^2 + b^2 + c^2 = 6$. Find the value of $a^3b + b^3c + c^3a$.

30. Let a, b, c be real numbers such that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a+b+c}.$$

Find the value of $(a+b)(b+c)(c+a)$.

31. (Finnish National High School Mathematics Competition 2002 Problem 2) Show that if $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a+b+c}$, then also

$$\frac{1}{a^n} + \frac{1}{b^n} + \frac{1}{c^n} = \frac{1}{a^n + b^n + c^n},$$

provided n is an odd positive integer.

32. Let $x_1, x_2, \dots, x_n$ be positive real numbers. Prove that

$$\frac{x_1x_3 - x_2^2}{(x_1 + x_2)(x_2 + x_3)} + \frac{x_2x_4 - x_3^2}{(x_2 + x_3)(x_3 + x_4)} + \dots + \frac{x_nx_2 - x_1^2}{(x_n + x_1)(x_1 + x_2)} = 0.$$

33. (IMO shortlist 1968) Find all solutions $(x_1, x_2, \dots, x_n)$ of the equation

$$1 + \frac{1}{x_1} + \frac{x_1 + 1}{x_1x_2} + \frac{(x_1 + 1)(x_2 + 1)}{x_1x_2x_3} + \dots + \frac{(x_1 + 1)(x_2 + 1) \cdots (x_{n-1} + 1)}{x_1x_2 \cdots x_n} = 0.$$

34. (CGMO 2012 Problem 1) Let n be a positive integer and $a_1, a_2, \dots, a_n$ nonnegative real numbers. Prove that

$$\frac{1}{1 + a_1} + \frac{a_1}{(1 + a_1)(1 + a_2)} + \dots + \frac{a_1a_2 \cdots a_{n-1}}{(1 + a_1)(1 + a_2) \cdots (1 + a_n)} \leq 1.$$

Root and Vieta's Theorem

35. Let α, β, γ be the roots of the equation $2x^3 - 9x^2 + x + 11 = 0$. Find the values of the following:

(a) $\alpha^2 + \beta^2 + \gamma^2$;

(b) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$;

(c) $2\alpha^3 - 6\alpha^2 + \alpha + 3\beta^2 + 3\gamma^2$.

36. Let α, β, γ be the roots of the equation $x^3 - 3x^2 - x + 2 = 0$. Find the monic cubic polynomial whose roots are $\alpha\beta, \beta\gamma, \gamma\alpha$.

37. (IMO shortlist 1966) Let $ax^3 + bx^2 + cx + d$ be a polynomial with integer coefficients a, b, c, d such that ad is an odd number and bc is an even number. Prove that (at least) one root of the polynomial is irrational.

(Remark: We may assume all roots of the polynomial are real. Otherwise, the conclusion may not hold.)

38. Let a, b, c be integers such that the roots of the equation $x^3 + ax^2 + bx + c = 0$ are exactly a, b, c . Find all possibilities of the triples (a, b, c) .
39. (USAMO 1973 Problem 4) Determine all roots, real or complex, of the system of simultaneous equations

$$\begin{aligned}x + y + z &= 3, \\x^2 + y^2 + z^2 &= 3, \\x^3 + y^3 + z^3 &= 3.\end{aligned}$$

40. (IMO shortlist 1971) Knowing that the system

$$\begin{aligned}x + y + z &= 3, \\x^3 + y^3 + z^3 &= 15, \\x^4 + y^4 + z^4 &= 35,\end{aligned}$$

has a real solution x, y, z for which $x^2 + y^2 + z^2 < 10$, find the value of $x^5 + y^5 + z^5$ for that solution.

41. (IMO shortlist 1982) Let $p(x)$ be a cubic polynomial with integer coefficients with leading coefficient 1 and with one of its roots equal to the product of the other two. Show that $2p(-1)$ is a multiple of $p(1) + p(-1) - 2(1 + p(0))$.
42. Let a, b, c be positive real numbers. If the polynomial $x^3 - ax^2 + bx - c = 0$ has three real roots, prove that all the roots are positive.
43. Let $(x_n)_{n \geq 1}$ and $(y_n)_{n \geq 1}$ be two infinite arithmetic sequences of real numbers with nonzero common differences. If $P(x)$ is a polynomial such that $P(x_n) = y_n$ for infinitely many n 's, prove that $P(x)$ is a linear polynomial
44. Let $P(x)$ be a polynomial. Prove that only finitely many real numbers x may satisfy $P(x) = \sqrt{x}$.
45. Prove that the equation $x^3 + 4x^2 + x - 1 = 0$ has three distinct real roots.
46. Prove that the equation $x(x-2)(x-4)(x-6) + (x-1)(x-3)(x-5)(x-7)$ has four distinct real roots.

47. (IMO shortlist 1968) If a_i ($i = 1, 2, \dots, n$) are distinct nonzero real numbers, prove that the equation

$$\frac{a_1}{a_1 - x} + \frac{a_2}{a_2 - x} + \dots + \frac{a_n}{a_n - x} = n$$

has at least $n - 1$ real roots.

48. Let $P(x)$ be a nonzero polynomial with real coefficients such that $P(x)$ divides the polynomial $P(x^2 + x + 1)$. Prove that $P(x)$ has no real roots.

Record

v1.0: problems 1–48